Skills Summary

Reading Arrays: Rows, Groups, and the Total

Use this reference while completing your worksheet or when studying.

Skill 1 — Read an Array as Multiplication

The model: an array is equal groups. Each *row* is one group.

rows × how many in each row = total

- "how many rows" → count down the side (the number of groups)
- "how many in each row" → count across one row (the size of a group)
- "how many in all / altogether / total" → multiply the two

Example: An array has 4 rows with 3 counters in each row. How many counters in all?
Step 1. "In all" asks for every counter, not for the number of rows, so this is the multiply question.
Step 2. Skip-count by threes, once for each of the 4 rows: 3, 6, 9, 12. So $4 \times 3 = \mathbf{12}$.

Try it: An array has 5 rows with 2 counters in each row. How many counters in all?

check: **10**

Skill 2 — Fix a Rows-Versus-Total Mistake

Two checks that catch the mistake every time:

- The total counts *objects*. The number of rows counts *groups*. They are different things.
- Unless every row holds exactly one object, the total is bigger than the number of rows.

Adding is not the fix either: rows + in each row answers nothing in the picture.

Example: Sara has 6 rows with 4 beads in each row. She says, "6 beads in all."
Step 1. Sara reported the number of rows; the question asked for beads, so she answered the wrong one of the two counts.
Step 2. Keep her two numbers and multiply: skip-count by fours six times — 4, 8, 12, 16, 20, 24. So $6 \times 4 = \mathbf{24}$.

Try it: Ben has 3 rows with 7 marbles in each row. He says "3 marbles." What is the correct total?

check: **21**

Skill 3 — Find the Missing Rows or Group Size

When the total is given and one part is missing:

rows × in each row = total → "_____ groups of the known size make the total"

Skip-count by the number you know and stop at the total. The number of counts you said is the missing number. Then check by multiplying back.

Example: 20 pencils are put into rows of 5. How many rows?
Step 1. The total (20) and the group size (5) are known, so the missing number is how many groups — that is the rows.
Step 2. Skip-count by fives to 20: 5, 10, 15, 20 is 4 counts. Check $5 \times 4 = 20$ ✓, so $\boldsymbol{n = 4}$.

Try it: 42 stickers are put into 6 equal rows. How many stickers are in each row?

check: $\boldsymbol{n = 7}$

(!) Watch out: Before you write an answer, say out loud what it counts — "rows" or "objects." If the question said *in all* and your answer is the row count, you answered the other question.